Design and Implementation of an Enhanced Point of Sale and Inventory Management System for Jolens Bicycle Supply

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***Abstract*—This paper discusses the development of an enhanced Point of Sale (POS) system tailored for Jolen's Bicycle Supply located in Digos City, Philippines. The new system was designed to tackle operational difficulties stemming from manual and paper-based procedures, which include inefficient sales transactions, inaccuracies in inventory tracking, and challenges in data management. The POS system streamlines core business tasks such as recording sales, updating inventory, and generating reports, which in turn improves both efficiency and accuracy in daily operations. Some of the worthy features of the system are real-time monitoring of inventory, centralized database management, and auto-summarization of data, with export capabilities of daily, weekly, and monthly reports. The system uses React.js for programming and MySQL as its backend database. Extensive testing was done in different environments to guarantee functionality as well as reliability. The use of this POS system is anticipated to greatly improve operational efficiency and accuracy, leading to improved decision-making for Jolen's Bicycle Supply.**

***Index Terms*—Point of Sale, inventory management, transaction automation, report generation, React, Node.js, MySQL**

I. Introduction

In today's business environment, the use of technology in the organization process of a system is an important aspect of maintaining highly efficient and accurate business operations. The introduction and subsequent usage of digital resources and systems have revolutionized the processes of businesses and companies, adding a new field to its competitiveness. As such, those medium sized and larger businesses can struggle with inefficiencies, errors, and slower decisions. Those in retail, business are readily accepting computer systems to handle sales tracking and inventory management to enhance workflow and user service experience.

A. Nature of the System and Company

Jolen's Bicycle Supply, which was founded in 2003 in Digos City, is a sole proprietorship engaged in the retailing of bicycles, spare parts, and cycling accessories. Its customers include children, hobbyists, and professional cyclists.

Jolen's Bicycle Supply currently utilizes a hybrid system that integrates manual and simple digital processes to run its day-to-day functions. Though this system has been working for the company all these years, it is now reflecting the signs of inefficiency, inaccuracy, and scalability issues. The current setup utilizes a lot of manual record-keeping with occasional Excel spreadsheet usage, which has caused operational bottlenecks, especially during seasons of high activity. Following is a comprehensive description of the major processes and their respective current workflows.

- **Inventory Management**
  - Manual inventory tracking through paper logs and Excel spreadsheets.
  - Monthly physical counts of stock and manual updates of discrepancies.
  - Manual determination of reorder points results in delays and inaccuracies.

- **Sales Transactions**
  - Manual entry of sales transactions from receipts and logbooks.
  - Lack of Excel skills among employees causes errors and data entry and analysis difficulties.
  - Manual processes make it difficult to monitor real-time sales performance.

This project is an extension of a previously created bicycle shop management system that offered backend API to employees, customers, suppliers, and inventory. Although that iteration established a foundation, its absence of a POS system, constraint of a limited interface, and overstretching of scope (to include non-essential modules like attendance tracking) made it not ready for real time business operation.

The planned version 2 POS System will concentrate on fundamentals: **sale transactions**, **inventory**, and **reporting**. It will have a new React frontend and keep the Node.js/MySQL backend with optimized database structures and less complexity to meet the shop's actual requirements.

B. Problem Statement

Jolen's Bicycle Supply experiences certain day-to-day functional issues with a paper-dependent system of recordkeeping. These challenges include **inaccurate inventory tracking**, **time-consuming sales processes**, and the **risk of data loss or damage**. Employees must manually check stock levels, which often results in errors and inefficiencies, while paper records are prone to being misplaced or destroyed. Consultation with the business owner and watching employees helped us pinpoint key issues of

Jolen's Bicycle Supply's system in use. These are inhibiting the business and complicating day-to-day operation more than they must. Here's what we determined:

- **Risk of Damage or Loss:** As things are today, the records are all on paper—sales, stock, and even purchase orders from suppliers. Paper is easily damaged, lost, or mislaid so important records are gone in a flash.
- **Time-Consuming and Messy:** It takes a lot of space and time to store and manage all these paper records. Employees waste a lot of time searching for material they need, particularly on hectic days.
- **Errors and Inconsistencies:** Writing things on paper by hand is prone to errors. Stock levels might not be current or sales amounts incorrectly recorded. Such mistakes can lead to overstocking, stockouts, and disappointed customers.

Sales transactions, currently taking **3–5 minutes**, might be simplified via a **digital system**, enhancing overall efficiency and security. These problems have their root causes in overdependence on manual methods, limited technical expertise among employees, and lack of data accuracy. These issues impact employees, management, and customers daily, particularly busy stockroom and cash station business hours. Transaction logging in a manual manner also hampers the retrieval or analysis of data in real-time. For instance, a customer inquiring after a particular bicycle model might receive inaccurate stock availability information through outdated records, resulting in sales losses and dissatisfied customers. To resolve such problems, a Point of Sale (POS) system implementation would greatly benefit inventory and sales management, enhance accuracy, decrease human errors, and streamline daily functions.

C. Objectives

The aim of this project is to develop a system to significantly **enhance the sales transactional efficiency and inventory administration** of Jolen's Bicycle Supply. Achieving this aim shall bring about streamlined business functions, a reduction in human mistakes, and higher client satisfaction. Aligning with the problem identified in the problem definition, the system objectives and major aims of this project are outlined as follows:

Project Goals

The aims of the study are the following in detail:

1) **Maximize Transaction Sales.** Improve transaction speed and accuracy, aiming to consistently achieve transaction times well below three minutes.
2) **Reduce Processing Time.** Automate sales and inventory tracking to minimize the time spent searching for product or record information.
3) **Data summarization.** Export summaries of processes of a day for historical analysis and storage.

System Objectives

1) **Automate Inventory Updates.** Product lists are automatically easy to maintain and view.
2) **Provide a Centralized Database.** Develop a secure and efficient database to systematically store and retrieve sales and inventory data.
3) **Generate Comprehensive Reports.** Implement automated report generation that offers daily, weekly, or monthly sales analysis.

D. Scope and Limitations

The project scope is centered on implementing an improved **Point of Sale** (POS) system specifically customized for the daily operating needs of Jolen's Bicycle Supply. The POS system **tracks**, **organizes**, and **facilitates** key business functions such as sales management, inventory tracking, and reporting. It allows for data table view and navigation, product addition and depletion from inventory, transactional processing, and daily sales summaries exportable for reporting. These capabilities provide a streamlined, efficient means of tracking inventory amounts, sales, and facilitating decision-making for daily business operations. A dashboard will provide a brief, high-level view of key performance indicators such as total sales, top-selling product lines, and low-stock notifications.

Nonetheless, the project does outline certain limits delimiting the areas it does not reach in its present form. Supplier management modules, reward schemes for customers, and attendance monitoring for staff are not included in favor of concentrating on the fundamental requirements of the shop. Additionally, no complex computing functionality such as pay-roll, tax, or forecasting is included in the system, as it is narrowly and specifically constrained in focus on the sales and inventory efficiency.

E. Review of Related Works

The evolution of computerized point-of-sale (POS) and inventory management systems was widely studied in different domains, offering a wealth of information applicable for Jolens Bicycle Supply's transition from manual to computerized processes. Such studies emphasize the aspect of user-oriented designs, customization for particular areas of application, inventory administration efficiency, software quality, and implementation feasibility in small retail.

Sidhunata et al. [1] used the Design Thinking approach in designing a POS system for a motorcycle repair shop. Its analysis underscores the necessity for designers to empathize with future system users and iteratively develop solutions tailored to practical workflow needs. Through digitized transaction recording, the system minimized human errors and enhanced business efficiency substantially. For Jolens, this strategy places particular emphasis on basing system design on actual user needs, given the limited familiarity of employees with computerized equipment.

In the bicycle industry, Subaktiono et al. [2] have proposed a web-based system for bicycle modification sales where customers can browse different components, request modifications, and make transactions over the internet. It includes features such as searching functions, electronic payment system integration, and customized recommendations. Such customized application indicates the possibility of web-dedicated systems in enhancing customer interaction and facilitating targeted bicycle services. For Jolens, the installation of similar web-driven functionalities can expand customer reach and service flexibility.

Dewangga [3] developed the design and prototype of an information system for Tamansari Bicycle Shop's warehouse with a focus on inventory tracking using barcodes. Paper methods were eliminated, and transaction as well as procurement times were minimized, directly tackling inefficiencies inherent in traditional shops. Its applicability to Jolens is important in the sense that digitized inventory tracking can enhance stock accuracy, avert shortages or

overstocking, and stock operational reliability in general.

From a system engineering standpoint, Rahman and Ripon [4] highlighted the significance of non-functional requirements (NFRs) in POS development. Employing UML use-case-based questionnaires, their survey categorized NFRs such as performance, security, reliability, and maintainability, claiming that neglecting such considerations in early stages of development renders systems delicate. For Jolens, rigorous identification of NFRs would guarantee the strength of its POS solution in favor of scalability and sustainability in the long run.

Finally, Lewis et al. [5] explored the feasibility of implementing mobile POS in small corner shops in Baltimore in a pilot experiment. It was determined to possess high operability and moderate feasibility with possibilities for further implementation in small shops. Although not quite bicycle-focused, the pilot gives an indication of technology deployment in small shops, and a tablet or a mobile POS system might be a suitable and low-cost business strategy for Jolens, as it is a small operation serving a localized clientele.

Individually, these studies offer a solid basis for Jolens Bicycle Supply's planned system. The literature focuses on user-centered designs [1], customized domains for bicycles [2], efficiency in operation through inventory in the digital space [3], system quality through non-functional requirements [4], and plausible models for adoption in small retail [5]. Collectively, they outline both technical and practical considerations needed to develop a POS and inventory system that is efficient, sustainable, and tailored to the particular needs of Jolens.

II. Software Development Methodology

The improvement of the advanced Point of Sale (POS) and Inventory Management System for Jolen's Bicycle Supply is carried out through an **iterative** and **incremental** approach. It was chosen because the project is built on a previously developed system (Version 1) in the Information Management course, which had already put in place basic functionalities for inventory and sales administration. In the present phase, the system is not redesigned from the absolute starting point; instead, it is being refined and extended (Version 2) to add a fully functional POS module, advanced reporting capabilities, streamlined database structures, and a modern user interface. The use of an iterative approach allows the team to bring in features in cycles of refinement. It begins with the gathering of feedback and requirements, then the designing and implementing of upgrades, and ends with testing and verification before moving on to the subsequent improvement. It assures that the system progresses in accordance with the real business needs of the shop but remains possible within the limitations of the academic timeframe.

In order to facilitate system design and promote clarity, the team utilized **Unified Modeling Language** (UML) diagrams. These diagrams serve to illustrate the functional parameters, workflows, interrelations among entities, user interactions, and the deployment context. Collectively, they constitute a framework that connects technical design with actual business operations, thereby ensuring that the final system is both feasible and resilient.

A. Use Case Diagram

Figure 1 shows the different use cases users will have when interacting with the program. There is an account creation and login system to access the rest of the program. For the inventory section an option to add or remove from the inventory, where these products can be selected in listing a transaction. After these transactions, the summary for the day is updated and can be viewed and exported as a file. And finally, users can view analytics from the data. This diagram is critical because it defines the functional boundaries of the system and ensures that the development team addresses the actual needs of end-users.

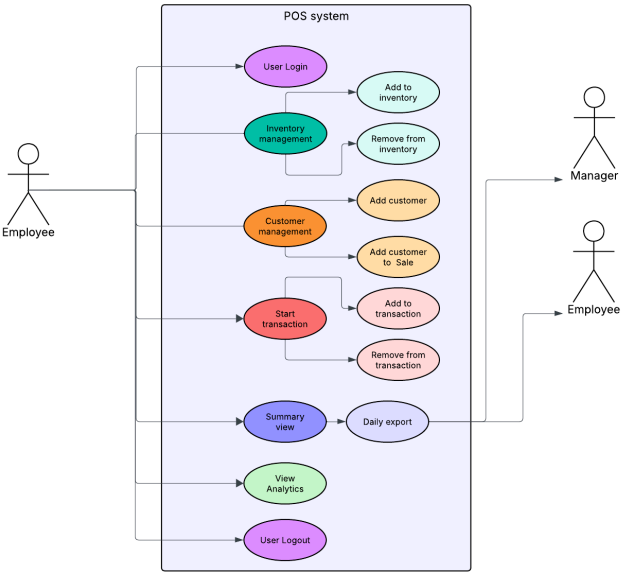


Fig. 1. Use Case Diagram for Jolens Pos and Inventory System

B. Activity Diagram

Figure 2 presents the overall flow of key processes within the system, covering inventory management, transactions, account access, data summaries, and analytics. It illustrates how users can perform operations such as adding or deleting items from the inventory, handling transactions by adding or removing items, and generating summaries for further analysis or export. It also shows supporting activities like importing summaries, logging in, and creating accounts, which ensure secure access and personalized use.

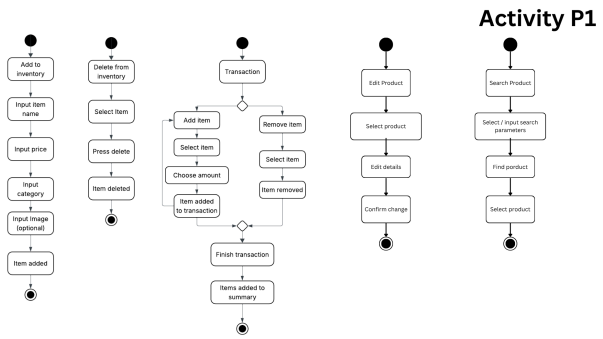


Fig. 2. Activity Diagrams of Jolens POS and Inventory System Part 1

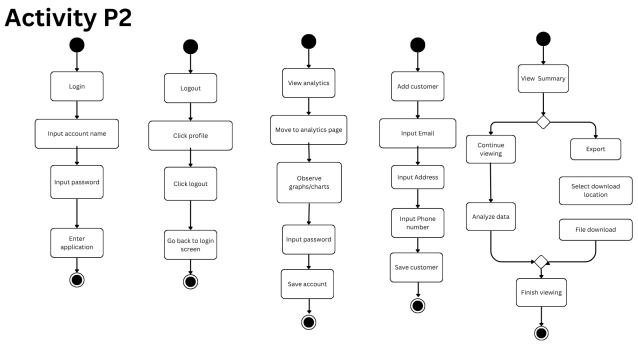


Fig. 3. Activity Diagrams of Jolens POS and Inventory System Part 2

C. *Class Diagram*

Figure 3 illustrates the structure of the application by defining its most significant classes, attributes, methods, and relationships among them. It illustrates how entities such as **accounts**, **products**, **customers**, **transactions**, **summaries**, and **sales** relate to each other to provide for the functionality of the system. Each class has some attributes that store key information and methods that define possible operations to make both data management and business processes account for. This diagram is a blueprint for developers, with a clear and organized model of the architecture of the system and that culminates in the implementation of its primary features.

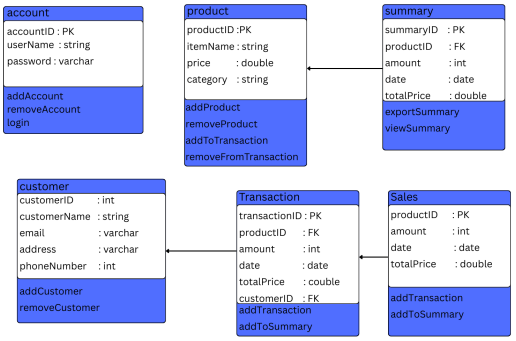


Fig. 4. Class Diagram for POS and Inventory Entities

D. *Deployment Diagram*

Figure 4 illustrates how system software components are distributed and communicate across the infrastructure. It illustrates the interdependence of core modules—analysis, transactions, inventory, summary, and authentication—and how they interface with the POS database and GUI. It identifies how system components are communicated among each other and data is handed across from the backend database to the front-end interface.

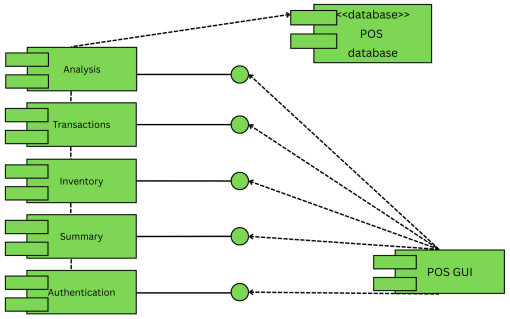


Fig. 5. Deployment Diagram of Jolens POS

and Inventory System

E. *Sequence Diagrams*

Figures 5-10 illustrate the step-by-step flow of interactions between users, the system interface, and backend services. They show the order in which processes occur, such as logging in, processing a transaction, updating stock, and generating a summary report. This provides clarity on the timing and dependencies of operations, reducing the likelihood of logical errors during coding.

Accounts

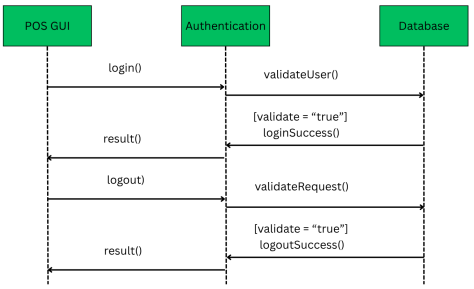


Fig. 6. Sequence Diagram for User Authentication (Accounts)

Transactions

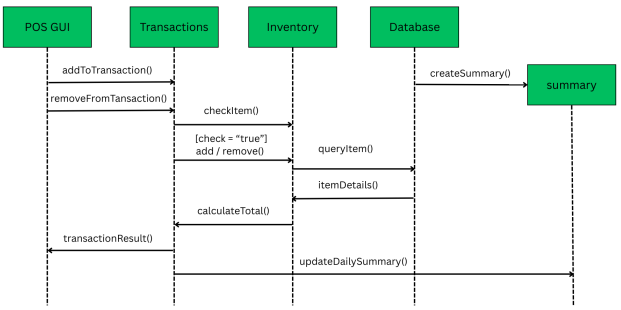


Fig. 7. Sequence Diagram for Sales Transaction Processing

Products

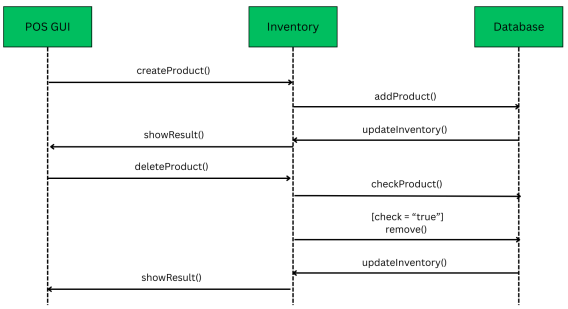


Fig. 8. Sequence Diagram for Product Management

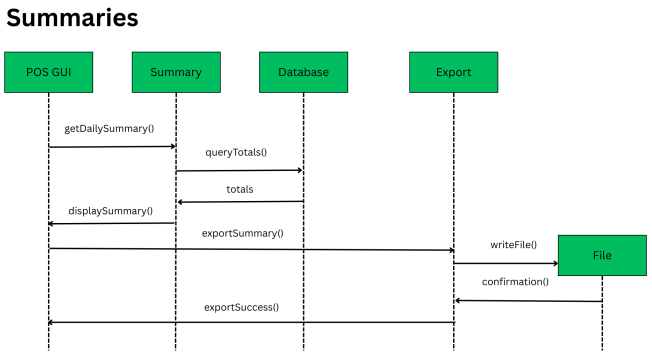


Fig. 9. Sequence Diagram for Report Generation (Summaries)

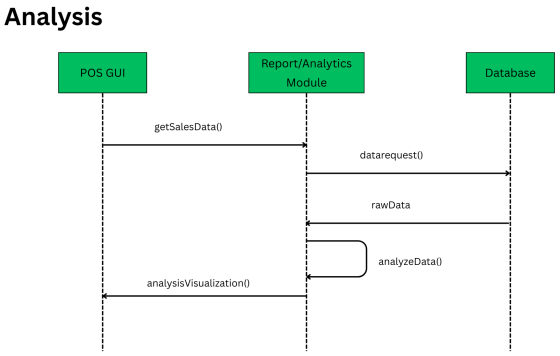


Fig. 10. Sequence Diagram for Analysis

III. Software Methodology and User Interface Design

The Bicycle Supply Management system is implemented as a **full-stack web application**, composed of a **backend REST API** and a **frontend single-page application (SPA)**. The architecture follows a **client-server model**, where the frontend interacts with the backend through HTTP REST endpoints. The backend relies on **Node.js, Express, and MySQL** (with Sequelize ORM), whereas the frontend uses **React, Vite, and Tailwind CSS** for a component-based, responsive interface. The separation of concerns ensures maintainability and scalability, while enabling real-time management of products, sales, employees, and stock.

3.1. Backend Architecture and Design

A. Overview

The backend serves as the **central data-processing and storage layer**, handling CRUD operations and domain-specific business logic for resources such as products, employees, sales, suppliers, and stock adjustments. Its primary responsibilities include data validation, transaction handling, authentication, and business-rule enforcement. The backend's modular structure allows controllers, routes, and models to be independently developed and tested, enhancing maintainability.

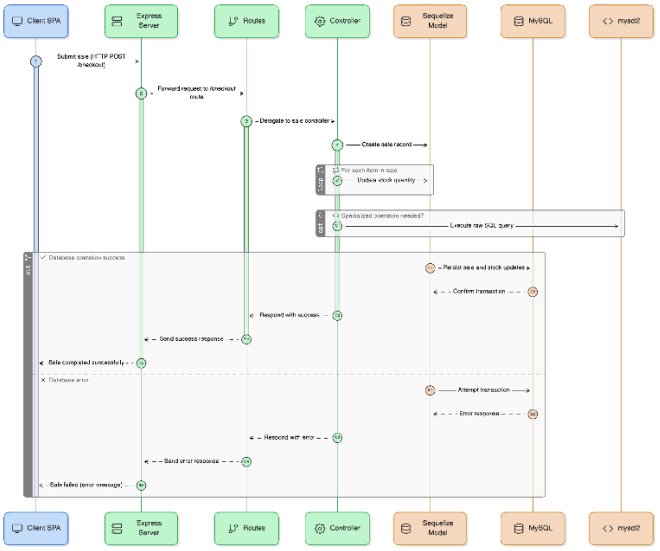


Fig. 11. Backend architecture: routes, controllers, models, DB

- **Project Root:** .../bicycle-supply-backend
- **Main entry point:** src/app.js – initializes Express, mounts middlewares, and route modules.
- **Database access:** MySQL through Sequelize ORM and optionally mysql2 for raw queries.

B. Technology Stack

The backend of the Bicycle Supply Management system employs a set of technologies chosen for efficiency, scalability, and maintainability. Node.js provides a server-side runtime environment using CommonJS modules, while Express.js manages routing and middleware for handling HTTP requests. MySQL serves as the relational database storing structured data, and Sequelize ORM maps models to database tables, simplifying query operations. Security is ensured with bcrypt, which hashes employee passwords before storage, and dotenv manages configuration of sensitive environment variables such as database credentials. CORS middleware enables safe cross-origin requests from the frontend, and nodemon improves development productivity by automatically restarting the server when file changes are detected. Table 1 below summarizes the technologies and their primary purposes.

Table 1. Backend Technology Stack and Their Roles

Technology	Purpose
Node.js	Server-side runtime environment (CommonJS modules)
Express.js	HTTP server framework managing routing and middleware
MySQL	Relational database system storing structured data
Sequelize ORM	Object-relational mapping for models and queries
bcrypt	Password hashing for employee authentication
dotenv	Configuration of environment variables
CORS	Middleware for cross-origin requests

nodemon	Development tool for auto-restarting server on file changes

C. File Structure and Components

The backend file structure promotes modularity:

- **src/config/db.js:** contains database configuration. Both Sequelize and mysql2 are configured, but the export currently overwrites the pool instance.
- **src/models/:** defines models for product, sale, employee, customer, stock_adjustment, etc., with proper associations (hasMany/belongsTo).
- **src/controllers/:** implement CRUD and business logic. Example: getProductQuantityOnHand computes current stock from supplies and sales.
- **src/routes/:** maps endpoints to controllers, mounted under /api/*.

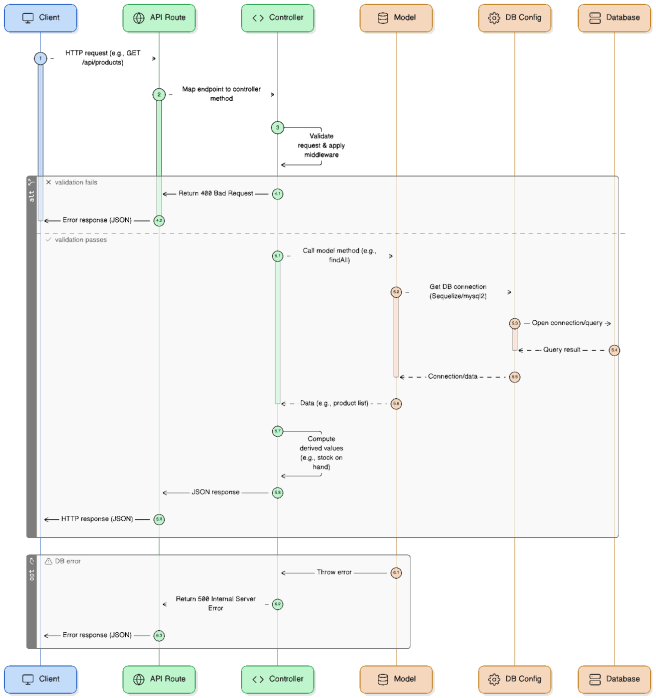


Fig. 12. Backend Module Interaction Diagram: Request Flow from Routes to Database

D. Request Flow

Requests follow a predictable sequence:

1. Client sends an HTTP request (e.g., POST /api/products).
2. Express routes the request to the appropriate controller.
3. Controller executes business logic and performs database operations via Sequelize.
4. JSON response is returned with the relevant status code.

E. Database Schema

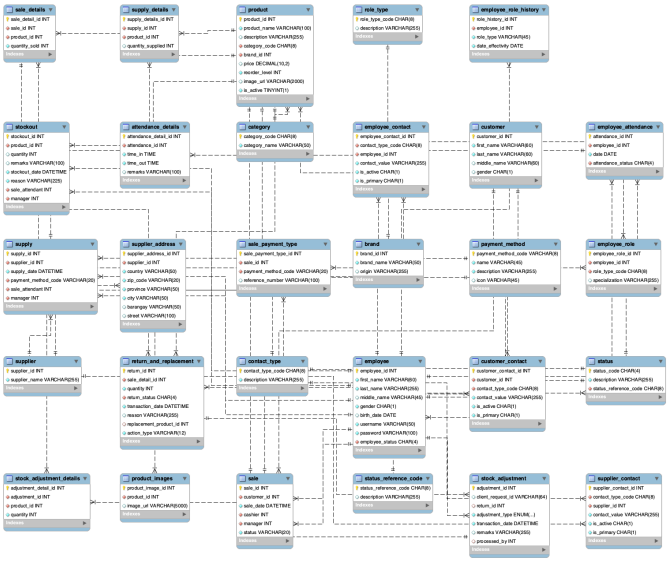


Fig. 13. Entity-Relationship Diagram (ERD)

The database employs a **normalized relational schema**:

- **Entities:** product, brand, category, sale, sale_details, supply, supply_details, employee, customer, stock_adjustment, status.
- **Stock is computed dynamically:** current_stock = SUM(quantity_supplied) - SUM(quantity_sold).
- This approach preserves **historical auditability** but may introduce performance overhead for large datasets.

F. Concurrency, Transactions, and Security

- Critical operations (sales, supplies, stock adjustments) require **transactions** to maintain data integrity.
- Authentication uses bcrypt, but session handling currently embeds passwords in tokens, which is insecure. JWT-based authentication is preferable.
- Input validation ensures robustness against invalid or malicious data.

3. 2. Frontend Architecture and Design

The frontend of the Enhanced Point of Sale and Inventory Management System for Jolen's Bicycle Supply is built as a **single-page application (SPA)** using **React.js**, styled with **Tailwind CSS**, and structured using **Vite** for optimized development and build processes. This approach enables a responsive, fast-loading, and modular interface that minimizes full-page reloads while providing a seamless user experience.

The application follows a **component-based architecture** where each feature (e.g., Point of Sale, Inventory Management, Reports) is designed as an independent module composed of reusable UI components. These components are logically grouped under /components/ and /screens/ directories, promoting scalability and code reuse.

Routing is handled through **React Router v6**, allowing role-based and authenticated navigation. The

AuthContext.jsx manages global user state, session persistence, and route protection for managerial functions. API communications are abstracted in a centralized api.js file, which interacts with backend REST endpoints (see Fig. 12) through **Axios**, ensuring consistency and maintainability across data-driven views.

- **Project Root:** .../bicycle-supply-frontend
- **Main entry:** src/index.jsx – mounts the app and context providers.
- **Routing:** Routes.jsx using React Router v6.
- **State management:** React Context (AuthContext.jsx) for authentication and session handling.

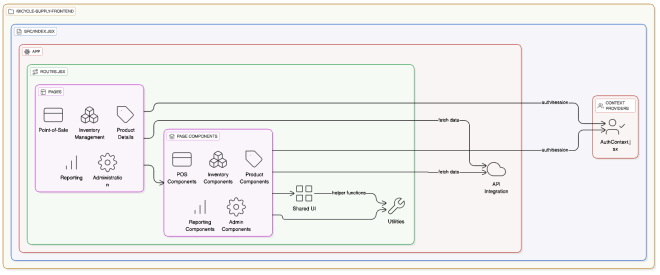


Fig. 14. Frontend Component Hierarchy

Fig. 14 illustrates the component hierarchy, showing the top-level structure that connects the navigation sidebar, context providers, and individual feature screens. The layered architecture ensures that the user interface (presentation layer), business logic (context and state), and data layer (API requests) remain separated, enhancing both performance and maintainability.

3. 3. User Interface

The user interface (UI) represents the visible component of the system, serving as the main avenue of engagement for the staff at Jolen's Bicycle Supply. The underlying design philosophy of this system emphasizes usability, simplicity, and efficiency. The aim is to have a clean and clear layout that lets staff with basic technical skills do important tasks with little training. By making a logical flow and a steady design, the interface acts as a guide, helping to save time on each transaction and reducing the frustration of dealing with a complicated and messy screen. Each screen is made to solve a specific business problem, like making checkout faster, improving inventory accuracy, and showing real-time data.

A. Dashboard

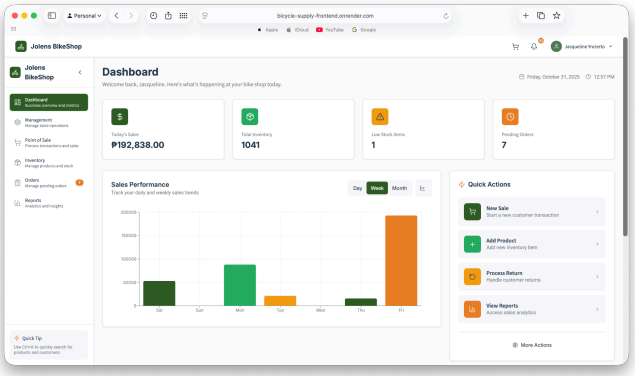


Fig. 15. The Main Dashboard

The Dashboard serves as the **main landing page** after user login, displaying a concise overview of store operations. As shown in Fig. 15, it summarizes daily and cumulative metrics such as total sales, top-performing products, and transaction counts.

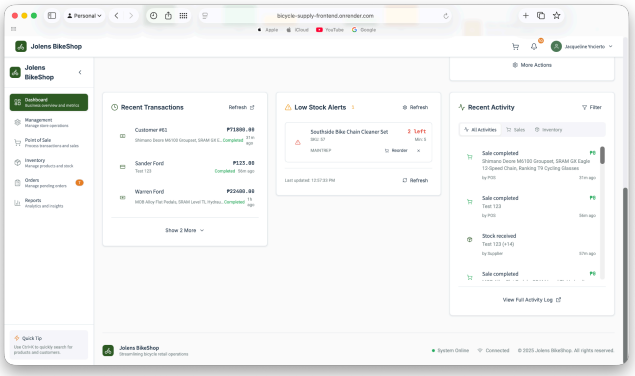
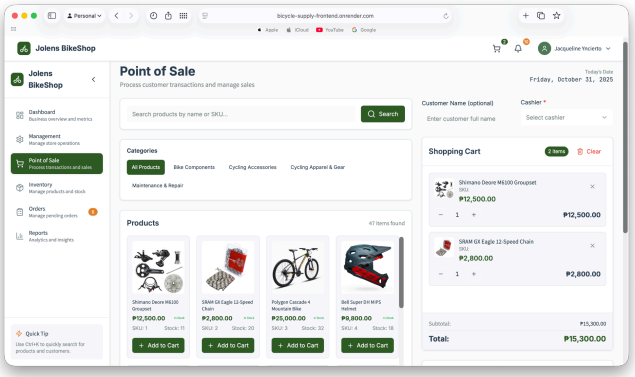


Fig. 16. Dashboard Recent Transactions, Low Stock Alerts, and Recent Activity Cards

Meanwhile, Fig. 16 presents supplementary cards for *recent transactions, low-stock alerts,* and *recent activity logs*, allowing staff to quickly identify urgent operational updates. The dashboard is dynamically populated via API calls that aggregate real-time data, offering immediate insight into business performance without manual report generation.

B. Point of Sale



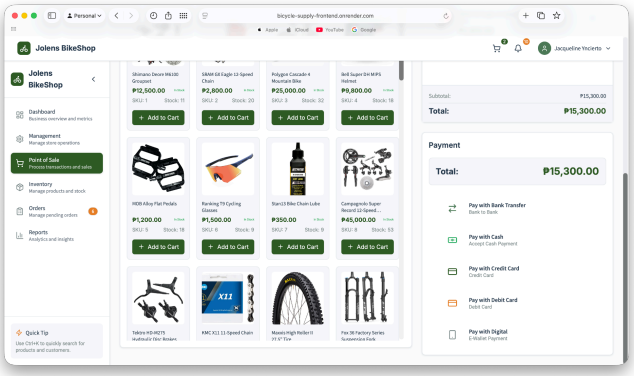
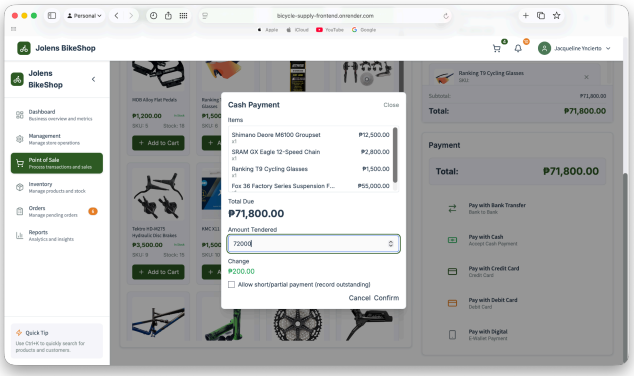


Fig. 17. The Main Sales Transaction

The POS interface, shown in Fig. 17, is designed for fast and intuitive transaction handling. Products can be searched or scanned, added to a dynamic cart, and quantities adjusted in real-time. The layout adopts a split-panel view — product listing on the left and cart summary on the right — enabling efficient multitasking.



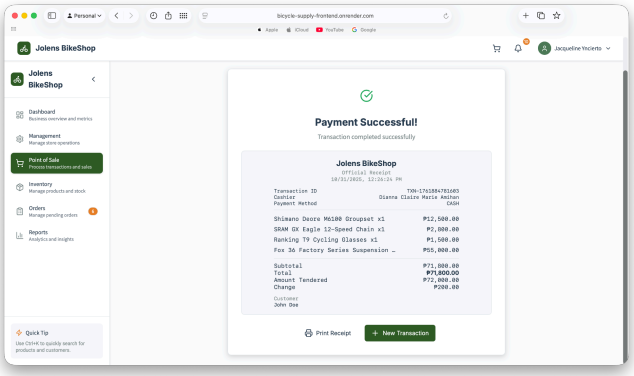


Fig. 18. Sales Transaction Payment and Payment Methods

Fig. 18 shows the payment processing interface, which supports multiple payment types (cash, GCash, card, etc.) and automatically calculates change and total due. Upon completion, each transaction triggers automatic updates to inventory quantities and generates a sales record in the database. The streamlined workflow reduces transaction time from minutes to seconds, improving cashier productivity.

Inventory Management

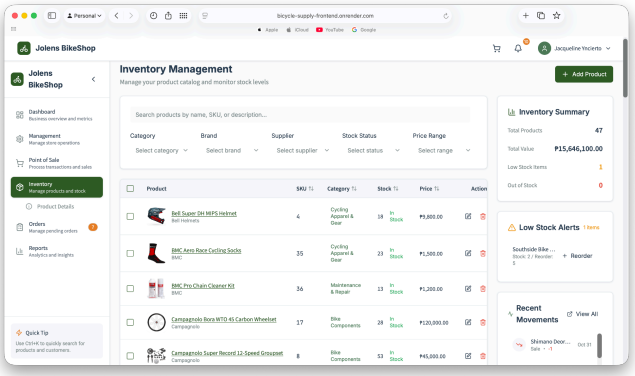


Fig. 19. The Inventory Management Table

Inventory Management is the backbone of the system, allowing real-time visibility of all stocked items. As seen in Fig. 19, the main inventory table lists all products with sortable columns and quick search filters.

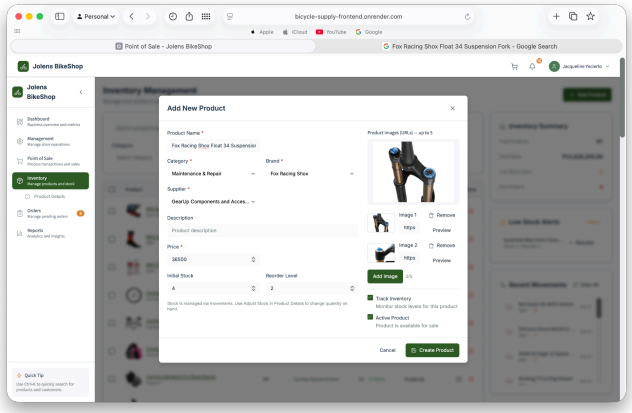


Fig. 20. Add New Product Modal

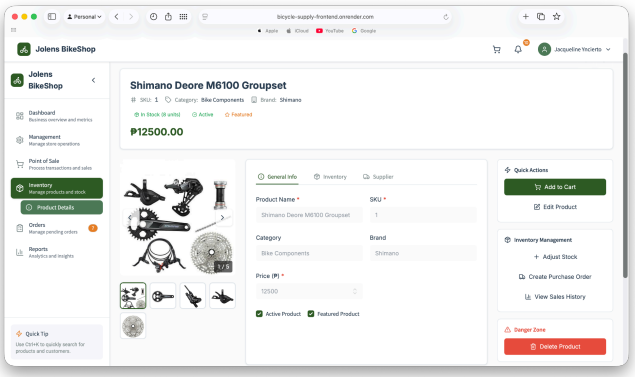


Fig. 21. Product Details General Info Tab

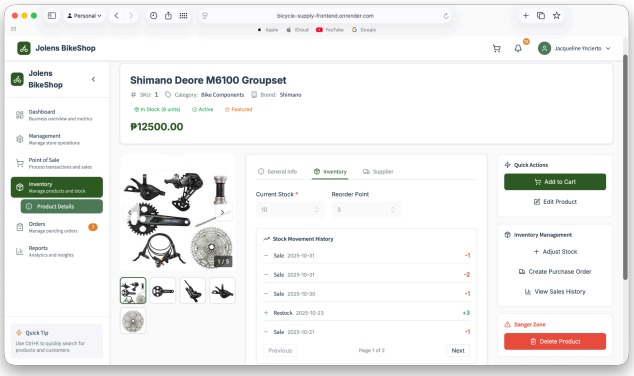


Fig. 22. Product Details Inventory Tab

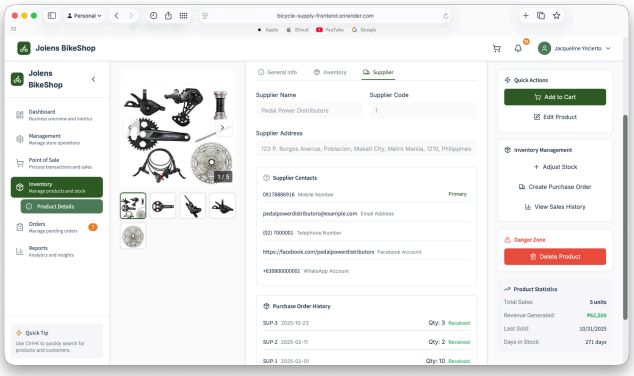


Fig. 23. Product Details Supplier Tab

Adding new products is accomplished via the modal interface in Fig. 20, while Figs. 21–23 detail the tabbed product information view, encompassing *General Info*, *Inventory Details*, and *Supplier Records*.

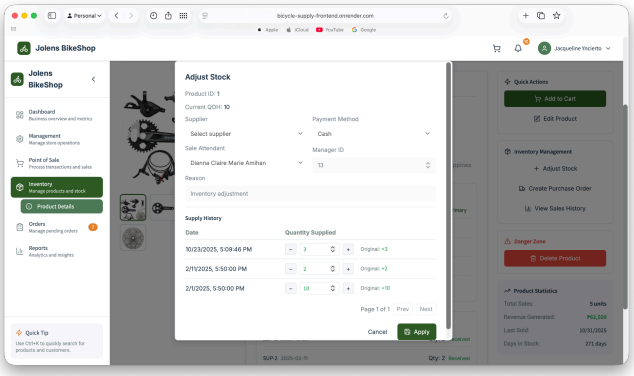


Fig. 24. Product Details Adjust Stock Modal

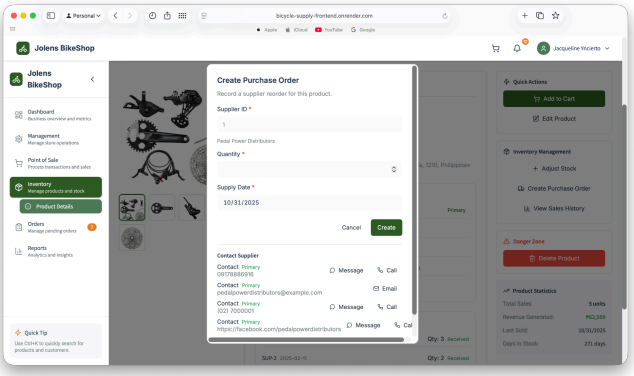


Fig. 25. Product Details Create Purchase Order Modal

Fig. 24 illustrates the *Adjust Stock* modal for quantity corrections due to damages or stock discrepancies, and Fig. 25 presents the *Purchase Order Creation* modal that records supplier restocks. Together, these interfaces ensure accurate and up-to-date stock levels, automating what used to be manual paper-based adjustments.

Sales Summary Report and Analytics

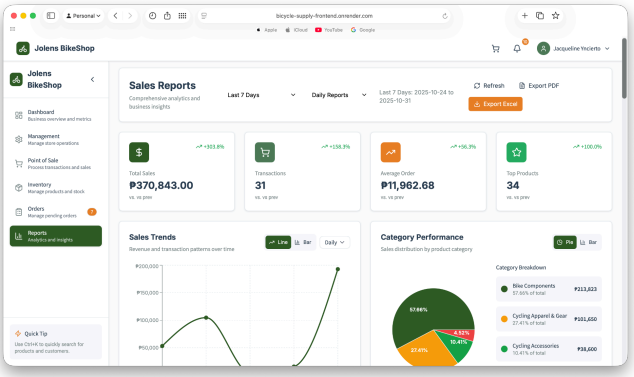


Fig. 26. The Sales Summary Report

The Sales Summary module provides in-depth analytics and transaction history visualization. Fig. 26 presents the main report dashboard with summarized totals and filters for date range selection.

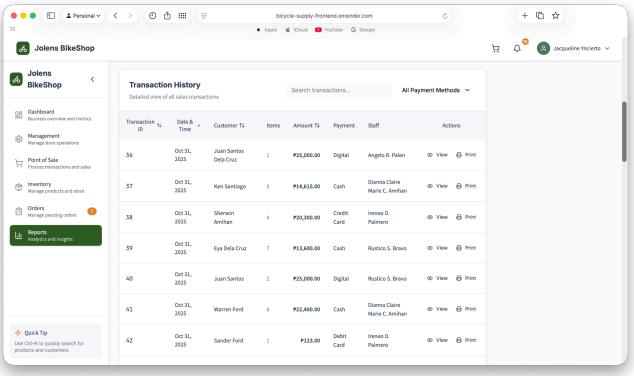


Fig. 27. The Sales Summary Report Transaction History Table

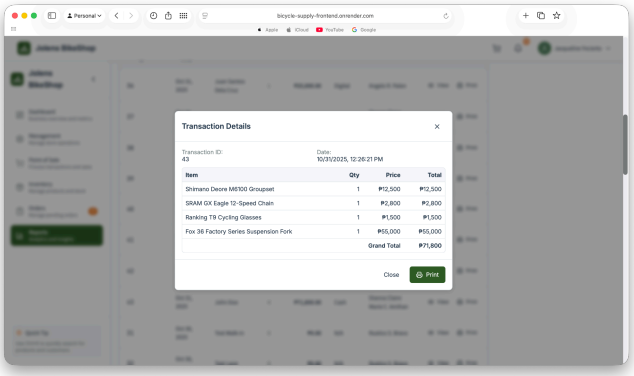


Fig. 28. The Sales Summary Report Transaction View Transaction Details

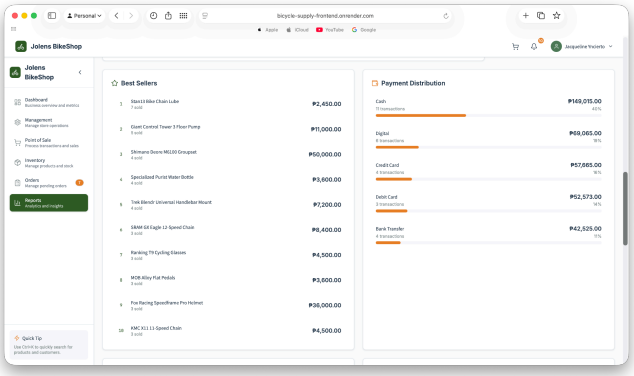


Fig. 29. The Sales Summary Report Best Sellers and Payment Distribution Widgets

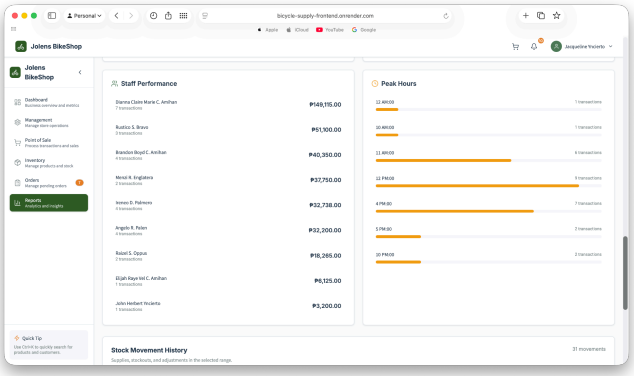


Fig. 30. The Sales Summary Report Best Performance and Peak Hours Widgets

Transaction history, displayed in Fig. 27, allows users to review past sales, while Fig. 28 offers detailed transaction breakdowns. Analytical widgets, shown in Figs. 29–30, include *best-selling products*, *payment distribution charts*, and *peak hour performance*, giving actionable insights into sales behavior.

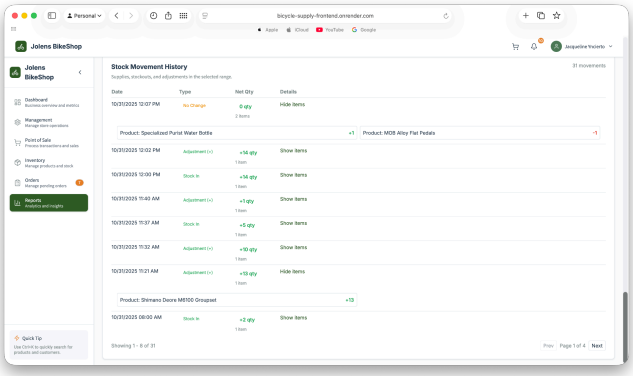


Fig. 31. The Sales Summary Report Stock Movement History Table

Lastly, Fig. 31 demonstrates *stock movement tracking*, where inflows and outflows are charted to aid restocking decisions. These analytic components are generated dynamically using aggregated backend data, reinforcing data-driven management.

Pending Orders

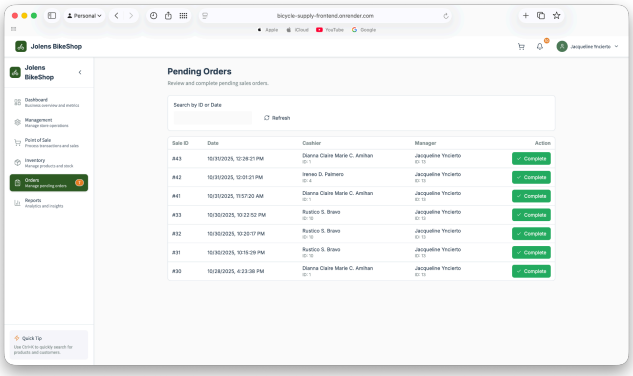


Fig. 32. Review and Complete Pending Orders

Managerial Control

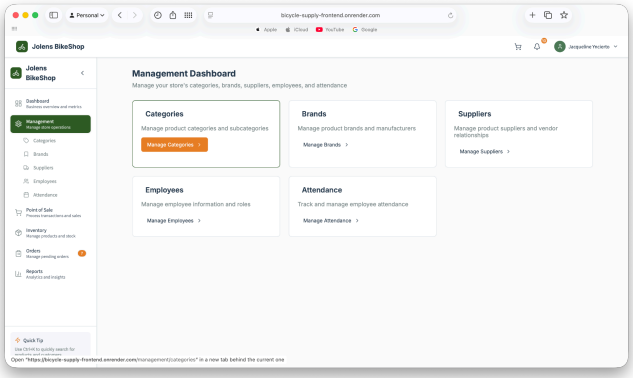


Fig. 33. Categories, Brands, Suppliers, Employees, and Attendance Management Dashboard

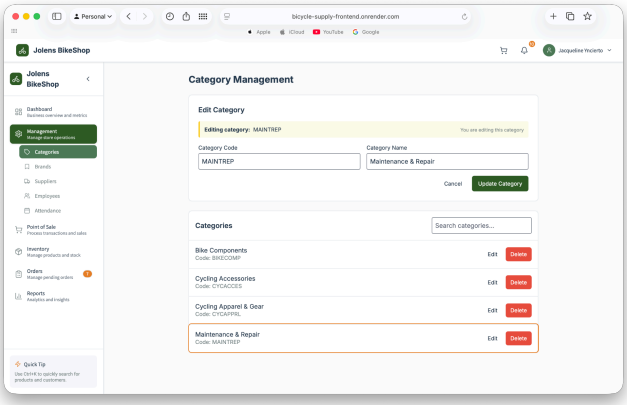


Fig. 34. Category Management

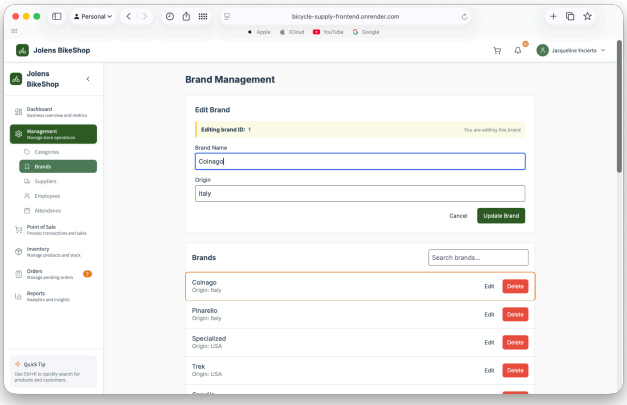
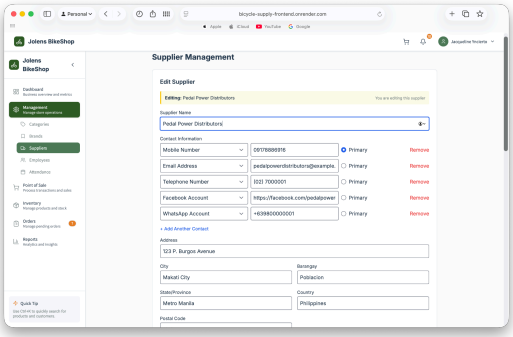


Fig. 35. Brand Management



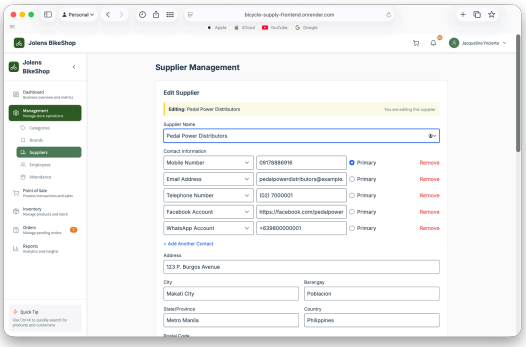


Fig. 36. Supplier Management

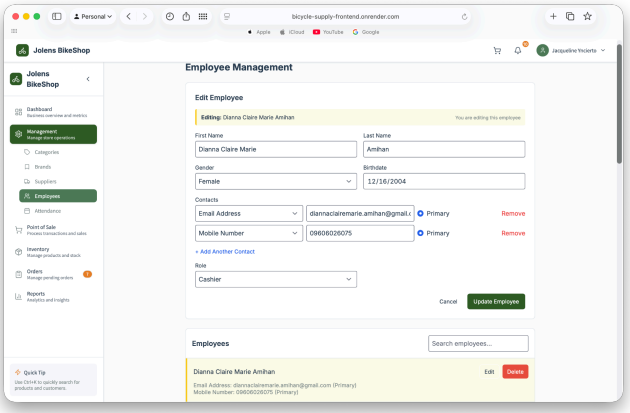


Fig. 37. Employee Management

Managerial Control encompasses administrative panels for managing categories, brands, suppliers, employees, and attendance. Fig. 33 shows the centralized management dashboard, while Figs. 34–37 demonstrate specific entity management screens with CRUD functionalities.

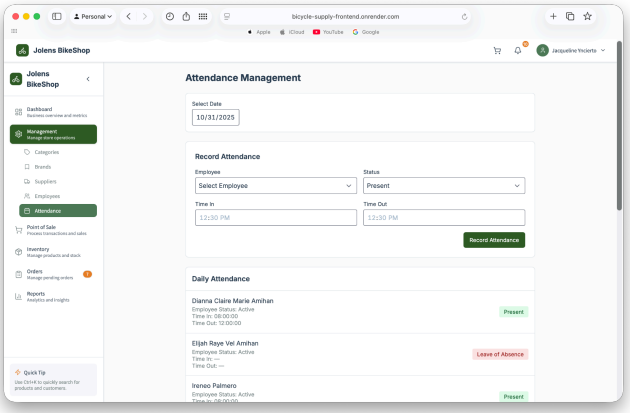


Fig. 38. Employee Attendance Management

The attendance monitoring interface in Fig. 38 provides visual logs of employee check-ins and work hours. These modules are protected by *manager-level authentication routes*, ensuring that sensitive business data and HR-related actions are securely managed by authorized personnel only.

Account Management and Settings

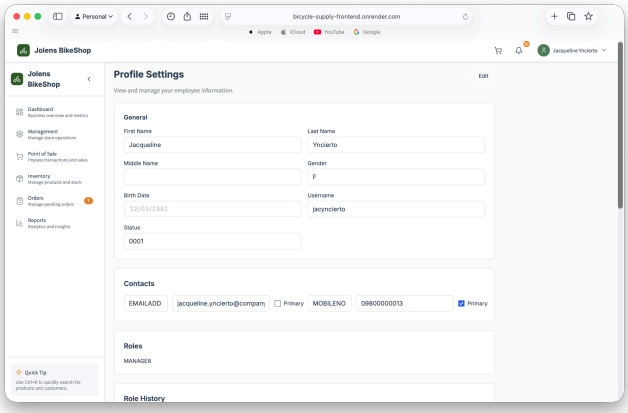


Fig. 39. Profile Settings

Account Management and Settings allow users to customize profiles, update credentials, and modify interface preferences. Fig. 39 shows the *Profile Settings* panel, where personal and role-based information can be edited.

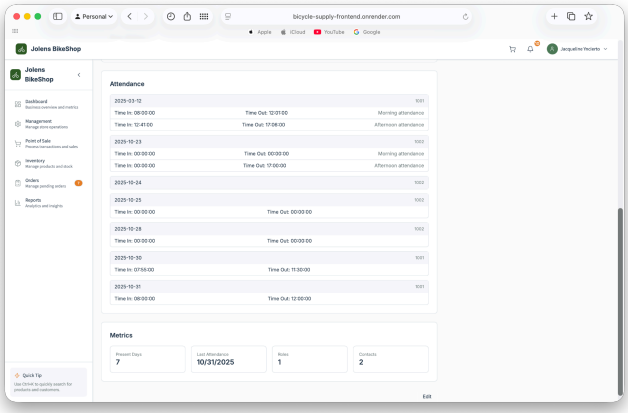


Fig. 40. Attendance Overview and Metrics

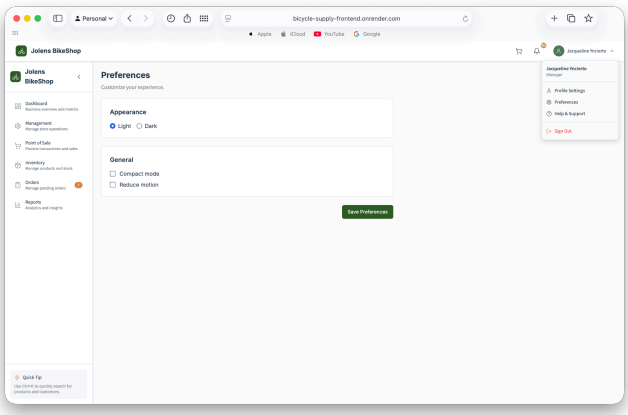


Fig. 41. Preference Settings

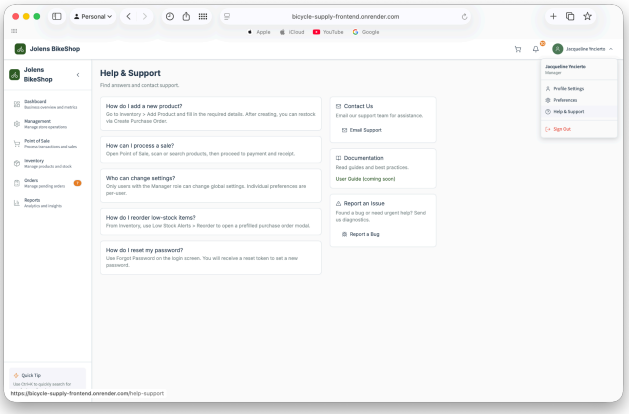


Fig. 42. Help & Support

Fig. 40 provides an overview of user attendance metrics, Fig. 41 shows configurable *Preference Settings* (such as display mode or report format), and Fig. 42 presents the *Help & Support* interface for quick access to guidance and technical assistance.

These sections enhance user autonomy, system usability, and operational transparency, aligning with the overall design philosophy of accessibility and reliability.

IV. Software Testing and Quality Assurance

A. Quality questionnaire

A questionnaire was created for the end users to answer after accessing the software. It consists of 28 questions categorized in the ISO/IEC 25010 standard for quality insurance. Google Forms was used to deploy the survey for ease of use and collection of data. Each question is multiple choice using a 5 point likert scale to determine the opinions of the tester on each question. The categories are:

1. Functional Suitability - Measures how well the system's features perform their intended tasks accurately and completely.

2. Performance Efficiency - Evaluates the system's speed, responsiveness, and ability to handle multiple operations effectively.

3. Compatibility - Assesses how well the system works with other software, databases, and file formats.

4. Usability - Determines how easy and intuitive the system is for users to learn, navigate, and operate.

5. Reliability - Checks if the system consistently performs as expected without crashes or data loss.

6. Security - Examines how well the system protects user data and prevents unauthorized access.

7. Maintainability - Measures how easy it is to update,

fix, or enhance the system without introducing new issues.

8. Portability - Assesses how easily the system can be installed, run, or used on different devices or environments.

After answering the questionnaire each question has its score averaged, and these scores are then used to determine the average score of the category they belong to. On a 5 point likert scale each score corresponds to a different rating:

1 – Strongly Disagree
2 – Disagree
3 – Neutral
4 – Agree
5 – Strongly Agree

10 end users tested the program and gave feedback through the google forms questionnaire. Below are the interpretation of the average score ranges and the summarized results of each category.

Table 2. Score Implementation

Score range	Interpretation
0 - 1	The software fails to meet quality requirements in this category; major issues or missing functionality.
1.1 - 2	The software partially meets expectations but still needs significant improvement in performance or reliability.
2.1 - 3	The software moderately satisfies requirements; acceptable but not consistently reliable or efficient.
3.1 - 4	The software meets most quality standards in this category; only minor issues are present.
4.1 - 5	The software fully satisfies or exceeds quality expectations; functions efficiently and reliably.

Table 3. Category Score

Category	Score
Functional Suitability	4.46
Performance Efficiency	4.25
Compatibility	4.03
Usability	4.33
Reliability	4.15
Security	4.07
Maintainability	3.8
Portability	4.4

Table 3 presents the average scores obtained for each software quality category based on the evaluation results. The findings indicate that the system performed exceptionally well across most categories, with all scores falling within the "Agree" to "Strongly Agree" range (3.1–5.0). The highest ratings were observed in Functional Suitability (4.46), Portability (4.40), and Usability (4.33), showing that the system effectively supports its intended functions, can be used easily by users, and operates well across different environments. Performance Efficiency (4.25) and Reliability (4.15) also received strong evaluations, suggesting that the system runs smoothly and consistently without major errors. Security (4.07) and Compatibility (4.03) were rated positively, indicating adequate protection of user data and integration with other modules or systems. Meanwhile, Maintainability (3.80), though slightly lower, still falls under the "Agree" range, implying that while the system is maintainable, there is room for improvement in ease of updates and debugging. Overall, the results suggest that the developed POS system meets the expected software quality standards and performs efficiently in supporting business operations.

B. Testing Pictures

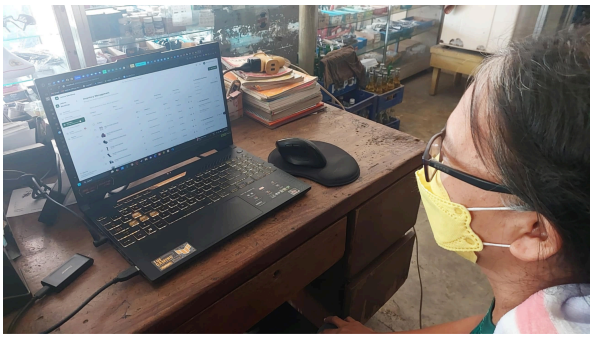


Fig. 42. Tester picture 1

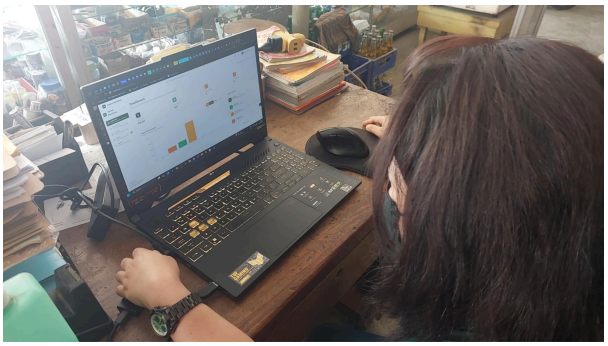


Fig. 43. Tester picture 2

C. Future implementations and suggestions

The development of the POS and inventory system for Jolen's Bicycle Supply covered the basic necessities and core functionalities expected to cover the same processes. Acting as a POS system that handles transactions as well as an inventory system that provides details on the stocks of the business, and a system that exports report summaries as a new process. However, there are some features that may be lacking or are yet to be implemented. One of the features to be implemented would be the following, a way for employees to

create an account, as of right now it would be added directly to the database rather through the GUI. And other features would be added through suggestions or new requirements from the end users and business owner.

V. Conclusion

The development of the Enhanced Point of Sale (POS) and Inventory Management System for Jolen's Bicycle Supply successfully addressed the inefficiencies of the shop's previous manual and paper-based processes. By integrating digital transaction recording, automated inventory updates, and report generation features, the system streamlined daily operations and improved data accuracy and accessibility. The use of React.js for the frontend and Node.js with MySQL for the backend provided a stable, scalable, and responsive framework suitable for small retail operations.

Through user testing and software quality evaluation based on ISO/IEC 25010 standards, the system demonstrated strong performance in functionality, usability, and portability, confirming its reliability and effectiveness in supporting business workflows. The results validate that the system meets the operational needs of Jolen's Bicycle Supply and provides a foundation for future scalability.

For future development, the team envisions expanding the system's accessibility through a dedicated online platform, improving account management features, and incorporating enhanced analytics and reporting tools. These enhancements will further improve efficiency, user experience, and adaptability, ensuring that the POS system continues to evolve with the growing demands of the business.

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